A UV-Finite Quantum Field Theory from Event-Density Primitives

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Abstract

We show that the Event-Density (ED) primitive stack yields a *structurally complete, UV-finite* quantum field framework when extended from single-chain dynamics (Arc R) and chain-mass structure (Arc M) to multi-chain field-level content (Arc Q). Three headline structural theorems carry the load. **Theorem Q1** (GRH unconditional FORCED): the gauge connection A_μ forced by Stage R.1 minimal coupling is the participation measure of a specific Case-P $(1/2, 1/2)$ gauge rule-type τ_g with gauge-invariant L3 interface, satisfying $\sigma_{\tau_g} = 0$ via the gauge-masslessness mechanism. All five GRH refinements (lightlike worldlines, gauge-quotient individuation, vertex-anchored commitment, non-Abelian extension, vacuum/particle status) close across Arc Q sub-stages, promoting GRH to unconditional FORCED — ED's first "particle-class must exist" prediction. **Theorem Q2** (canonical (anti-)commutation FORCED): the standard QFT commutation relations $\{a_\tau^\dagger, a_\tau^\dagger\} = 0$ (Case R) and $[a_\tau^\dagger, a_\tau^\dagger] = 0$ (Case P) are forced directly from the spin–statistics theorem (Arc R) plus Primitive 10 individuation; second quantisation is derived, not postulated. **Theorem Q3** (UV-FIN FORCED at primitive level): Primitive 01 event-discreteness, Primitive 13 finite proper-time intervals, and Primitive 04 bounded bandwidth jointly guarantee primitive-level finiteness of all multi-chain participation integrals; renormalisation is REFUTED as a fundamental requirement (admissible only as continuum-approximation effective machinery), and the cosmological-constant divergence-form puzzle is structurally dissolved while Λ's magnitude remains INHERITED. Supporting structural results include the FORCED U(1) gauge structure with non-Abelian extensions admissible, the Fierz-basis vertex catalogue with mass-like $\bar{\Psi}\Psi A^2$ vertices REFUTED under unbroken gauge, multiple admissible Higgs-like SSB mechanisms, the linear-algebra-forced existence of mixing matrices and CP-violating phases under multi-generation conditions, and lightlike worldlines for $\sigma = 0$ rule-types via the Stage R.1 Casimir. Specific gauge group, coupling values, generation count, Higgs-mechanism choice, and Λ magnitude all remain INHERITED. ED yields the *form* of QFT structurally; numerical content is empirical.

1 Introduction

Quantum field theory (QFT) is built on a layered set of independent postulates: canonical (anti-)commutation relations, gauge invariance as a fundamental symmetry principle, the Higgs mechanism for spontaneous symmetry breaking, renormalisation procedures for ultraviolet divergences, and a fixed gauge group SU(3) × SU(2) × U(1) with specific matter content. The framework's empirical success is unparalleled, but its conceptual foundations involve substantial structural inputs not derived from a more fundamental ontology.

The Event-Density (ED) program offers a structural alternative. Phase-1 [1] derived non-relativistic single-particle quantum mechanics from $P_K = \sqrt{b_K}\, e^{i\pi_K}$ + four-band decomposition + commitment dynamics. Arc R [2] extended this to the relativistic regime, deriving Klein–Gordon, the spin–statistics theorem, the Cl(3,1) Clifford-algebra frame, and the Dirac equation. Arc M [3] addressed the chain-mass problem, establishing that ED forces the *form* of mass content via

the bandwidth-signature σ_τ but inherits all numerical mass values, ratios, and hierarchies as empirical content.

Arc Q, the subject of the present paper, completes the quantum sector by extending ED to the multi-chain, field-level, interaction-level QFT regime. Eight sub-stages address the major structural targets: gauge-group structure (Q.2), interaction vertices (Q.3), Higgs-like SSB (Q.4), radiative corrections (Q.5), generations and flavour (Q.6), second quantisation (Q.7), and vacuum structure (Q.8). The opening Q.1 evaluates the gauge-field-as-rule-type hypothesis (GRH).

Three Arc Q theorems extend the framework substantively:

- **Theorem Q1 (GRH unconditional forced).** ED's first "particle-class must exist" prediction.
- **Theorem Q2 (canonical (anti-)commutation forced).** Second quantisation derived from Arc R spin–statistics + Primitive 10 individuation.
- **Theorem Q3 (UV-FIN forced at primitive level).** ED is UV-complete without renormalisation; cosmological-constant divergence-form puzzle structurally dissolved.

UV-FIN is the genuinely novel structural prediction of Arc Q. It reframes the long-standing UV-divergence problems of QFT as continuum-approximation artefacts rather than fundamental pathologies, with implications for the cosmological-constant problem and high-energy phenomenology near the primitive event-discreteness scale.

The methodological framing established in Arc M — *form* FORCED, *value* INHERITED — extends to Arc Q at greater scope. ED's primitive structure determines the *form* of QFT (gauge structure, vertex catalogue, second quantisation, vacuum, UV behaviour) while inheriting numerical content (gauge group, couplings, masses, generations, Higgs parameters, Λ).

2 Background

2.1 ED primitives relevant to QFT

- **Primitive 01 (micro-event):** discrete events on the 3+1-dimensional event manifold.
- **Primitive 02 (chain):** worldlines γ_K.
- **Primitive 04 (bandwidth):** four-band decomposition with bounded amplitudes per primitive event.
- **Primitive 06 (four-gradient):** Lorentz covariance.
- **Primitive 07 (rule-type):** levers L1–L4.
- **Primitive 10 (individuation):** threshold separating distinct chains.
- **Primitive 11 (commitment):** discrete events along γ_K.
- **Primitive 13 (relational timing):** proper time τ_K.

2.2 Rule-type ontology

Spin–statistics from Arc R [2] connects L4 to L2 via $\eta = (-1)^{2s}$:

- **Case R** ($\eta = -1$, half-integer spin, fermionic). Bilinear: $|\bar{\Psi}\Gamma_\tau\Psi|$.
- **Case P** ($\eta = +1$, integer spin, bosonic). Bilinear: $|\Psi|^2$.

The **gauge rule-type** τ_g is a Case-P rule-type carrying the $(1/2, 1/2)$ Lorentz representation with gauge-invariant L3 interface.

2.3 Multi-chain field structure

The Phase-1 participation measure extends to multi-chain content via

$$\Phi_\tau(x) = \sum_{K \in \tau} P_{K,\tau}(x). \tag{1}$$

This is the natural extension of Phase-1's coherent-sum to relativistic multi-rule-type content.

3 Arc Q Stage Q.0 — Scoping

The Arc Q opening memo [4] frames eight structural targets (Q.1 through Q.8) and records an honest verdict prior of 40% mixed / 30% mostly-positive / 20% Arc-M-style negative-dominant / 10% surprise. Arc Q closes in the **30% mostly-positive quadrant**, with UV-FIN's promotion to FORCED constituting the closest thing to a structural surprise.

4 Arc Q Stage Q.1 — GRH Evaluation

4.1 GRH formal statement

The gauge-field-as-rule-type hypothesis posits that A_μ from Stage R.1 minimal coupling is the participation measure of a specific rule-type τ_g satisfying:

- **(GRH-1)** Case P (bosonic, integer spin).
- **(GRH-2)** $(1/2, 1/2)$ Lorentz representation (four-vector).
- **(GRH-3)** L3 interface enforces local gauge invariance.
- **(GRH-4)** $\sigma_{\tau_g} = 0$ via MR-P (gauge-masslessness, Arc M [3]).

4.2 Q.1 verdict

Stage Q.1 [5] verdict: **CANDIDATE-STRONG REQUIRES REFINEMENT**. All four clauses primitive-compatible; no clause REFUTED. Five refinements identified for closure:

- **R-1.** Lightlike-worldline reformulation.
- **R-2.** Gauge-quotient individuation.
- **R-3.** Vertex-anchored commitment.
- **R-4.** Non-Abelian extension.
- **R-5.** Vacuum/particle status.

Each refinement is closed across Q.2–Q.8.

5 Arc Q Stage Q.2 — Gauge-Group Admissibility

5.1 U(1) FORCED at primitive level

Stage R.1's local-phase invariance forces the U(1) gauge structure with covariant derivative $D_\mu = \partial_\mu + (iq/\hbar)A_\mu$.

5.2 Non-Abelian extension admissible

Stage Q.2 [6] establishes that any compact Lie group with finite-dimensional adjoint is structurally admissible. **The specific Standard-Model gauge group** $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ **is not forced** — confirmed EMPIRICAL.

5.3 R-2 closure: gauge-quotient individuation

Stage Q.2 closes R-2 via the **adjacency-equivalence-class construction**: individuation is defined on equivalence classes $[A_\mu]$ under gauge transformations.

5.4 CC-U1 (charge quantisation) CANDIDATE

A new Q.2 CANDIDATE: U(1) compactness $\Rightarrow$ wavefunction-phase periodicity in 2π $\Rightarrow$ charge quantisation. Plausible from Primitive 09; remains CANDIDATE.

6 Arc Q Stage Q.3 — Vertex Catalogue

6.1 Fierz-basis classification

Vertex type	Lorentz	Gauge	Admissibility
$\bar{\Psi}\gamma^\mu\Psi A_\mu$	scalar	invariant	FORCED (R.1/R.3)
$\bar{\Psi}\gamma^\mu\gamma^5\Psi A_\mu$	scalar	invariant	ADMISSIBLE
$\bar{\Psi}\sigma^{\mu\nu}\Psi F_{\mu\nu}$	scalar	invariant	ADMISSIBLE (effective)
$\bar{\Psi}\gamma^5\Psi\cdot$ pseudo-A	scalar	invariant	ADMISSIBLE
$\bar{\Psi}\Psi A_\mu A^\mu$	scalar	NOT invariant	REFUTED (unbroken gauge)

The REFUTED mass-like vertex becomes the structural target of Stage Q.4 (Higgs SSB).

6.2 R-4 closure: non-Abelian minimal coupling

Stage Q.3 [7] establishes that the gauge-covariant derivative

$$D_\mu\Psi = (\partial_\mu + igA^a_\mu T^a)\Psi \tag{2}$$

is the unique linear connection making $D_\mu\Psi$ transform covariantly under local SU(N), with A^a_μ transforming as

$$A^a_\mu \to A^a_\mu + \frac{1}{g}\partial_\mu\alpha^a - f^{abc}\alpha^b A^c_\mu. \tag{3}$$

This closes R-4.

6.3 Non-Abelian self-couplings

The field strength

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^c_\nu \tag{4}$$

contains f^{abc} commutator terms producing AAA + AAAA self-interactions. FORCED conditional on SU(N) admittance.

6.4 R-3 closure: vertex-anchored commitment

Stage Q.3 closes R-3: each commitment event of τ_g occurs at an interaction vertex with at least one charged-matter rule-type. Between vertices, τ_g propagates lightlike with no commitment events.

6.5 Forbidden structures

REFUTED:

- Gauge-non-invariant vertices under unbroken gauge.
- Non-local vertices (Primitive 11 locality).
- Lorentz-non-scalar combinations.
- Anyonic-statistics vertices in 3+1D.

7 Arc Q Stage Q.4 — Higgs-Like SSB

7.1 Five candidate mechanisms

Stage Q.4 [8]:

- **H1 scalar-rule-type Higgs τ_H:** ADMISSIBLE-CLEAN.
- **H2 bandwidth-shift via spatially-patterned condensate:** CANDIDATE.
- **H3 composite $\bar{\Psi}\Psi$ condensate:** ADMISSIBLE-EFFECTIVE.
- **H4 gauge-fixing artefact:** REFUTED.
- **H5 vacuum-anchored:** REFUTED as distinct (reduces to H1/H2).

7.2 Notable structural finding: H2 in naive form fails

Arc M's σ_τ master formula uses log-derivatives $\partial_\mu \ln b$, which are immune to uniform amplitude shifts $b \to \alpha b$. Therefore a uniform Higgs-VEV-like condensate alone cannot produce mass via H2 in naive form. H2 requires **spatially-patterned** condensate $\delta b(x)$. This is a non-trivial primitive-level constraint emerging from Arc M [3].

7.3 Verdict

ED admits multiple structurally-clean SSB routes; specific Higgs mechanism is EMPIRICAL inheritance.

8 Arc Q Stage Q.5 — Radiative Corrections

8.1 Loop structure form-level admissible

Stage Q.5 [9] establishes that loop-level QFT phenomena have primitive-level analogues via multi-chain commitment dynamics. Numerical coefficients are uniformly INHERITED.

8.2 Vacuum-polarisation transversality FORCED

GRH-3 forces $q_\mu \Pi^{\mu\nu}(q) = 0$ structurally.

8.3 Tree-level $g = 2$ preserved; only $\sigma^{\mu\nu}$ contributes to a_μ

FORCED: only the tensor $\sigma^{\mu\nu}$ Fierz bilinear contributes to anomalous magnetic moment. Axial / pseudoscalar bilinears contribute only to electric dipole moment (CP-violating).

8.4 UV-FIN CANDIDATE opened

Stage Q.5 opens UV-FIN: Primitive 01 event-discreteness suggests primitive-level multi-chain participation integrals are finite. Promoted to CANDIDATE-STRONG at Q.7; promoted to FORCED at Q.8.

9 Arc Q Stage Q.6 — Generations and Mixing

9.1 Six generation mechanisms evaluated

Stage Q.6 [10]:

- **G1 rule-type duplication:** ADMISSIBLE-CLEAN.
- **G2 bandweight-pattern:** REFUTED (collapses to G1).
- **G3 Fierz-class:** REFUTED (type not generation).
- **G4 vacuum-anchored:** REFUTED as mass-differentiation; admissible-as-labelling only.
- **G5 radiative splitting:** REFUTED phenomenologically.
- **G6 no-generation null:** ADMISSIBLE default.

Generation count = 3 is refuted-as-forced; confirmed empirical.

9.2 Linear-algebra theorems for mixing and CP

Two structural existence claims survive:

- **Mixing matrices forced to exist** under multi-generation + non-diagonal Yukawa (linear-algebra theorem).
- **CP phases forced to exist** under complex Yukawa + ≥ 3 generations.

Specific values INHERITED.

10 Arc Q Stages Q.7 & Q.8 — Second Quantisation and Vacuum Structure

10.1 Multi-chain field structure

Stage Q.7 [11] establishes

$$\Phi_\tau(x) = \sum_{K\in\tau} P_{K,\tau}(x) \qquad \text{FORCED.} \tag{5}$$

The vacuum $|0\rangle_\tau$ is the no-commitment configuration. Multi-rule-type vacuum factorises:

$$|0\rangle_{\text{full}} = \bigotimes_\tau |0\rangle_\tau. \tag{6}$$

10.2 Creation/annihilation events

Chain-creation and -annihilation events correspond to participation amplitudes crossing the Primitive 10 individuation threshold. These are the primitive-level realisations of QFT creation/annihilation operators, derived from Primitives 01 + 10 + 11.

10.3 Canonical (anti-)commutation FORCED

For Case-R, two chain-creation events at coincident points are forbidden by Primitive 10:

$$\{a_\tau^\dagger(x), a_\tau^\dagger(x')\}\big|_{x=x'} = 0 \qquad \text{(Case R)}. \tag{7}$$

For Case-P, coincidence is admissible:

$$[a_\tau^\dagger(x), a_\tau^\dagger(x')] = 0 \qquad \text{(Case P)}. \tag{8}$$

forced.

10.4 R-1 closure: lightlike worldlines for $\sigma = 0$

Stage Q.7 closes R-1 via the dispersion-relation argument: $\sigma = 0 \Rightarrow$ Stage R.1 Casimir $P^2 = 0 \Rightarrow$ null worldline. Lightlike-affine-parameter λ replaces proper time. **R-1 closed.**

10.5 R-5 closure (Stage Q.8): vacuum/particle dual structure

Stage Q.8 [12] closes R-5:

- **Strict vacuum** $|0\rangle_\tau^{\text{strict}}$: no committed chains.
- **Effective vacuum** $|0\rangle_\tau^{\text{eff}}$: includes transient τ-participation in b^{env} of other rule-types.

Both well-defined at primitive level. Lorentz + gauge invariance preserved. **R-5 closed.**

10.6 GRH unconditional FORCED

With all five GRH refinements closed, GRH **promotes from CANDIDATE-STRONG to unconditional forced.** Back-flow to Arc M [3]: F-M8 promotes from conditional to unconditional FORCED — ED's first "particle-class must exist" prediction.

10.7 UV-FIN PROMOTED TO FORCED

Three-step argument:

1. Each primitive-level configuration is finite (Primitives 01 + 04).
2. Finitely many configurations per region (Primitives 01 + 13).
3. Bounded total amplitude (Primitive 04 + flux-continuity).

Therefore: any primitive-level multi-chain participation integral is finite. **Renormalisation refuted as fundamental requirement.** Cosmological-constant divergence-form structurally dissolved; magnitude INHERITED.

10.8 G4 final REFUTED; H5 reduces to H1/H2

- G4 REFUTED as mass-differentiation; admissible-as-labelling only.
- H5 REFUTED as distinct mechanism; reduces to H1 or H2.

11 Structural Theorems

11.1 Theorem Q1 (GRH unconditional FORCED)

Theorem 11.1 (GRH unconditional FORCED, Q1)**.** *The gauge connection A_μ forced by Stage R.1 minimal coupling is the participation measure of a specific Case-P $(1/2, 1/2)$ gauge rule-type τ_g with gauge-invariant L3 interface, satisfying $\sigma_{\tau_g} = 0$ via the gauge-masslessness mechanism MR-P. All four GRH clauses are structurally* FORCED *at primitive level. The existence of at least one massless Case-P rule-type is therefore* FORCED *— ED's first "particle-class must exist" prediction.*

Sketch. GRH evaluation at Q.1 establishes CANDIDATE-STRONG with five refinements. Each closes:

- **R-1** closed at Q.7 via $\sigma = 0 \Rightarrow P^2 = 0 \Rightarrow$ null worldline.
- **R-2** closed at Q.2 via adjacency-equivalence-class on $[A_\mu]$.
- **R-3** closed at Q.3 via Primitive 11 commitment-locality at vertices.
- **R-4** closed at Q.3 via gauge-covariant-derivative uniqueness.
- **R-5** closed at Q.8 via dual vacuum-particle structure.

With all five refinements closed, GRH promotes to unconditional FORCED. F-M8 back-flow promotion follows. □

Dependencies on primitives. Stage R.1; Primitive 07 L3; Stage R.2.2; R.2.5 spin–statistics; Arc M MR-P; Primitives 01–13 jointly via the five refinement closures.

11.2 Theorem Q2 (Canonical (anti-)commutation FORCED)

Theorem 11.2 (Canonical (anti-)commutation FORCED, Q2)**.** *The canonical (anti-)commutation relations of QFT,*

$$\{a_\tau^\dagger(x), a_\tau^\dagger(x')\}\big|_{x=x'} = 0 \quad \textit{(Case R)}, \tag{9}$$

$$[a_\tau^\dagger(x), a_\tau^\dagger(x')]\big|_{x=x'} = 0 \quad \textit{(Case P)}, \tag{10}$$

are FORCED *at primitive level by Arc R's spin–statistics theorem $\eta = (-1)^{2s}$ combined with Primitive 10 individuation. Second quantisation is derived, not postulated.*

Sketch. Define chain-creation events as participation-amplitude threshold-crossings (Primitive 10). The corresponding operator-level construction $a_\tau^\dagger(x)$ inherits two properties from Arc R [2] R.2.5:

(i) For Case R ($\eta = -1$, vanishing-on-coincidence), two chain-creation events at the same event point are excluded by Primitive 10.

(ii) For Case P ($\eta = +1$, coincidence permitted), two chain-creation events at the same point are admissible.

Vanishing-on-coincidence translates to anticommutation; coincidence-admissibility translates to commutation. These are the canonical QFT (anti-)commutation relations. □

Dependencies on primitives. Arc R R.2.5 (spin–statistics); Primitive 10 (individuation); Primitive 01 (event-manifold structure); Primitive 11 (commitment events).

11.3 Theorem Q3 (UV-FIN FORCED at primitive level)

Theorem 11.3 (UV-FIN FORCED, Q3)**.** *Primitive 01 event-discreteness, Primitive 13 finite proper-time intervals, and Primitive 04 bounded bandwidth jointly guarantee that all multi-chain participation integrals are finite at primitive level. UV divergences in continuum QFT are continuum-approximation artefacts. Renormalisation is* REFUTED *as a fundamental requirement; admissible only as continuum-approximation effective machinery. The cosmological-constant divergence-form puzzle is structurally dissolved while* Λ*'s magnitude remains* INHERITED.

Sketch. Consider an arbitrary multi-chain participation integral $\mathcal{A}$. In the continuum approximation:

$$\mathcal{A}_{\text{cont}} = \int d^4k_1\, d^4k_2 \cdots \text{(integrand)}. \tag{11}$$

In ED's primitive-level realisation:

$$\mathcal{A}_{\text{ED}} = \sum_{\text{discrete configurations}} \text{(primitive-level amplitudes)}. \tag{12}$$

Three-step argument:

1. Each summand is finite: a primitive-level configuration involves a finite number of micro-events (Primitive 01) with bounded amplitudes (Primitive 04).
2. Configuration count is finite per region: Primitive 01 discreteness + Primitive 13 finite intervals.
3. Total amplitude bounded: Primitive 04 conservation.

Therefore $\mathcal{A}_{\text{ED}}$ is finite for any finite spacetime region. UV divergences of the conventional form $\int d^4k\, k^2 \to \infty$ *do not arise structurally* — they are artefacts of the continuum integral over-extending the primitive-level domain. The cosmological-constant naive QFT calculation $\Lambda \sim M^4_{\text{Planck}}$ is artefactual; primitive-level Λ is finite. Magnitude remains INHERITED. □

Dependencies on primitives. Primitive 01 (event-manifold discreteness); Primitive 04 (bounded bandwidth); Primitive 13 (finite proper-time intervals); Primitive 11 (commitment-event structure); Stage Q.7 second-quantisation framework.

12 Discussion

12.1 What Arc Q forces vs what it leaves empirical

forced at primitive level (Arc Q): GRH (Theorem Q1); canonical (anti-)commutation (Theorem Q2); UV-FIN (Theorem Q3); U(1) gauge structure; vector vertex; non-Abelian D_μ + AAA/AAAA conditional on SU(N); vertex locality + gauge-invariance filter; vacuum-polarisation transversality; default gauge masslessness; Goldstone absorption + photon survival under partial SSB; mixing matrices + CP phases existence; lightlike worldlines for $\sigma = 0$; multi-chain field structure; vacuum factorisation; effective-vacuum content; cosmological-Λ admissible-as-finite; tree-level $g = 2$; only $\sigma^{\mu\nu}$ contributes to a_μ.

inherited (not predicted): specific gauge group; numerical couplings; charge quantisation pattern; specific Higgs mechanism + parameters; Yukawa values; generation count; CKM/PMNS elements; CP phase values; mass values, ratios, hierarchies; cosmological-Λ magnitude; specific zero-point magnitudes; β-function coefficients.

refuted as forced: SM gauge group as forced; mass-like $\bar{\Psi}\Psi A^2$ under unbroken gauge; non-local / non-Lorentz-scalar / anyonic vertices; specific Higgs mechanism forced; generation count = 3 forced; renormalisation as fundamental; specific Λ value.

12.2 Why ED yields a UV-finite QFT

UV finiteness is not an additional primitive but a consequence of Primitives 01, 04, 13 jointly. Each primitive is independently motivated by Phase-1 structural requirements; their joint consequence is automatic UV finiteness. This is structurally cleaner than standard QFT renormalisation but does not predict numerical magnitudes — the form of UV finiteness is FORCED, the values remain INHERITED.

12.3 Renormalisation as continuum-approximation artefact

Standard QFT renormalisation handles UV divergences arising from naive continuum integrals. Under UV-FIN, these divergences arise because the continuum integral over-extends the actual primitive-level domain. This does not invalidate renormalisation as a calculational tool; renormalisation is REFUTED only as a *fundamental* requirement.

12.4 Implications for the cosmological constant

The cosmological-constant problem reframes under UV-FIN:

- QFT $\Lambda \sim M^4_{\text{Planck}}$ is artefactual.
- Primitive-level Λ is finite, bounded by event-discreteness scale.
- "Why is the divergence cancelled to 10^{-122}?" is replaced by "what numerical structure determines the finite primitive-level value?"

ED does not predict 10^{-122}. Magnitude remains INHERITED. But the divergence-form puzzle is structurally dissolved.

12.5 Relationship to standard QFT and effective field theory

Arc Q's relationship to standard QFT:

- Form-level FORCED content matches standard QFT structure.
- Numerical content matches standard QFT only via empirical fit.
- UV behaviour differs: standard QFT requires renormalisation; ED forces primitive-level UV finiteness.
- Effective field theory is the natural continuum-approximation framework for ED at scales much larger than the event-discreteness scale.

12.6 Implications for cosmology and Phase-3

Phase-3 (ED $\rightarrow$ GR coupling, cosmological structure) is unblocked. Arc Q's Phase-3 contributions: UV-FIN reframing of cosmological-Λ; GRH unconditional FORCED provides structurally-grounded gauge-mediator class; UV-FIN empirical signatures could probe the divergence-cancellation reframing at high energies.

13 Conclusion

Arc Q completes the quantum sector of ED. Three structural theorems carry the load: GRH unconditional FORCED (Theorem Q1); canonical (anti-)commutation FORCED (Theorem Q2); UV-FIN FORCED (Theorem Q3).

Together with Phase-1, Arc R, and Arc M, Arc Q yields a **structurally complete, UV-finite quantum field framework**. The form of QFT is forced from primitives. Numerical content is inherited as empirical fit.

The methodological framing — *form* FORCED, *value* INHERITED — extends cleanly from Arc M to Arc Q at QFT scope. ED's primitive structure produces classifications, dichotomies, and form-relations; it does not produce continuous numerical relationships.

Phase-3 directions — ED to general-relativistic coupling, cosmological structure via the UV-FIN reframing of cosmological-Λ, possible high-energy empirical signatures near the primitive event-discreteness scale — are unblocked by the Phase-2 closure of which Arc Q is the QFT extension.

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